COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 DATA STRUCTURES AND ALGORITHMS	**SECTION:**	CPE21S3
DATE PERFORMED:	JULY 28 2026	**DATE SUBMITTED:**	AUGUST 1 2026
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ACTIVITY NO. 4.1_ HANDS-ON ACTIVITY: STACKS

1. PROCEDURE OUTPUT

ILO A:

HOA4p1 > ILOA > main.cpp > main()

```cpp
#include <iostream>
#include <stack> // call the stak from STL


int main(){
    std::stack<int> newStack; // created an object

    // testing if the stack is empty
    std::cout<<"is the stack empty? "<<newStack.empty()<<std::endl;

    // insert 3 elements:
    newStack.push(3);
    std::cout<<"Top of stack: "<<newStack.top()<<std::endl;
    newStack.push(2);
    std::cout<<"Top of stack: "<<newStack.top()<<std::endl;
    newStack.push(1);
    std::cout<<"Top of stack: "<<newStack.top()<<std::endl;

    std::cout<<"is the stack empty? "<<newStack.empty()<<std::endl;
    newStack.push(6);
    newStack.push(7);
    newStack.push(8);

    std::cout<<"Size of the stack is: "<<newStack.size()<<std::endl;
    //stop here for documentaion

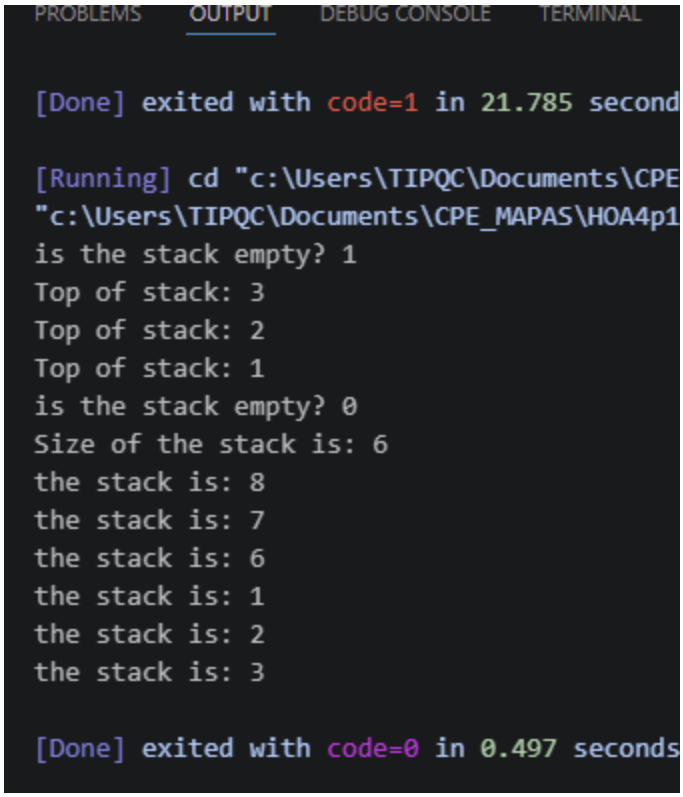
    //experimentation
    for( int i = 0; i<6; i++){
        std::cout<<"the stack is: "<<newStack.top()<<std::endl;
        newStack.pop();

    }

    //swap?? - swap the content of the public member functions
    //emplace?? - consruct and insert elements - public member function

    return 0;
}
```

Output:



Code analysis ILOA:
First off, I used iostream and stack library so that I can use the functions or methods for the stack. Next is the main function wherein I created an object called newStack which is a stack. Next is that I tested if the stack is empty by using the empty method, which will return a boolean value 1 because it is true that the stack is empty as of now. After that, I pushed 3 elements using the push method, and every push I displayed the element that is on the top of the stack with the top method. Next is that I checked if the stack is empty, which now returns false. After that, I pushed another three elements and then checked the size of the stack which is now six using the size method. After that, I did a little experimentation in which I displayed all the elements inside the stack using a for loop, and the top, and pop method.

ILOB:

HOA4p1 > ILOA > ILOB > main.cpp > main()

```cpp
#include <iostream>

const size_t maxCap = 100;
int stack[maxCap];
int top = -1, newData, elements;

//prototyping
void push();
void pop();
void Top();
bool isFull();
bool isEmpty();
void displayAll();


int main(){
    int choice;
    //asked the user for the maximun elements o be inserted
    std::cout<<"enter the number of maximun elements for the new sack (max: 100) ";
    std::cin >> elements;

    while(choice != 7){
        std::cout<<"Stack Operations: "<<std::endl;
        std::cout<<"1. Push \n2. Pop \n3. Top\n4. isEmpty\n5. isFull \n6. displayAll\n7. Exit! \n"<<std::endl;
        std::cin>>choice;

        switch(choice){
            case 1: push();
            break;
            case 2: pop();
            break;
            case 3: Top();
            break;
            case 4: std::cout<<isEmpty()<<std::endl;
            break;
            case 5: std::cout<<isFull()<<std::endl;
            break;
            case 6: displayAll();
            break;
            case 7: break;
            default: std::cout<<" Invalid choice"<<std::endl;
        }
    }
    return 0;
}
```

```cpp
}
bool isEmpty(){
    //top is -1
    if(top == -1) return true;
    return false;
}

```

```
bool isFull(){
    //not equal to the maxCap
    if(top >= elements-1){
        std::cout<<"Stack Overflow ";
        return true;
    }else return false;

}
```

```
61   void push(){
62       //check if the stack is full, if yes urn an error
63       if(isFull()) return;
64       //ask the user for a value to be pushed ino the stack
65       std::cout<<"What number you want to add to the stack? ";
66       std::cin>> newData;
67       //store inside the stack
68       stack[++top] = newData;
69
70   }
71
72   void pop(){
73
74       //check if thre are no more elements insie the stack
75       if(isEmpty()) return;
76       // display the element to be popped;
77       std::cout<<"Popping: "<<stack[top];
78       //decrement top
79       top--;
80   }
81
82   void Top(){
83       //error checking
84       if(isEmpty()) return;
85       //display the top element
86       std::cout<<"The top element is: "<< stack[top]<<std::endl;
87   }
```

```cpp
void displayAll(){
    if(isEmpty()) return;
    std::cout<<"The elements are: "<<std::endl;
    for(int i = top; i >= 0; i--){
        std::cout<<stack[i]<<std::endl;
    }
}
```

Output:

```
Executing task: C:/Windows/System32/cmd.exe /d /c .\build\Debug\o

enter the number of maximun elements for the new sack (max: 100) 2
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. displayAll
7. Exit!

1
What number you want to add to the stack? 6
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. displayAll
7. Exit!

1
What number you want to add to the stack? 7
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. displayAll
7. Exit!

6
The elements are:
7
6
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. displayAll
7. Exit!

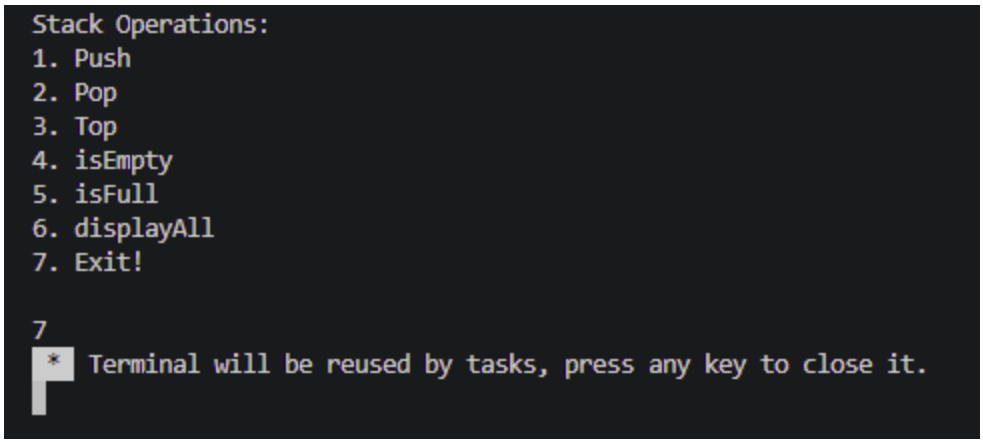
2
```

```
Popping: 7
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. displayAll
7. Exit!

6
The elements are:
6
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. displayAll
7. Exit!

4
0
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. displayAll
7. Exit!

5
0
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. displayAll
7. Exit!
```



Code Analysis ILO B:
First is that I defined size_ t maxCap to be equal to 100. After that, I defined a stack using a static array and where the top is equal to -1, newData, and elements. After that, I did my function prototyping. After that, inside the main function I declare choice as an integer, and then the program will ask what the maximum elements that the user wants to input to the program. Next is that I created a while loop with a condition that it will run if the user's choice is not 7. I then displayed the menu, and the user can input their choice. After that I did a switch case where every case has a designated function like push, pop, Top, isEmpty, isFull, displayAll, and Exit. The pop() function pops the top of the stack. The push() function pushes an element in the stack. The Top function displays the element that is on the top of the stack. The isEmpty() and isFull() functions checks if the stack is empty or full. And lastly the displayAll() function where it displays the element inside the stack starting from the top.

ILO B2:

Header File:

listStack.h > checker<T>()

```cpp
#ifndef LIST_STACK_H
#define LIST_STACK_H


template <typename T>
class Node{
    public:
    T data;
    Node<T>* next = nullptr;
};

//create a pointer/reference tip that alays pont to the current top node

//global pointer declaration

template <typename T>
Node<T>* head = nullptr;

//pushing operation:
template <typename T>
void push(T newData){
    //create a new node
    Node<T>* newNode = new Node<T>;
    //assign th data to the newnode
    newNode->data = newData;
    //point the nw node next to the current top
    newNode->next = head<T>;
    // assign the new top to the new node
    head<T> = newNode;
}

//isEmpty()
template <typename T>
bool isEmpty(){
    return head<T> == nullptr;
}
```

```cpp

template <typename T>
T pop(){
    // if the tack is empty error catch
    if(isEmpty<T>()){
        std::cout<<"Stack underflow"<<std::endl;
    }
    // saving a current top node to a temporary variable
    Node<T>* temp = head<T>;
    // store the data to a temporary variable
    T tempVal = temp->data;
    // move the head to next node
    head<T> = head<T>->next;
    //free up the memory
    delete temp;
    //return the temporry variable data
    return tempVal;
}

template <typename T>
void Top(){
    if(isEmpty<T>()) return;
    std::cout<<"The top is: "<<head<T>->data<<std::endl;
}

template <typename T>
void displayAll(){
    if(head<T> == nullptr) return;
    while(head<T> != nullptr){
        std::cout<<head<T>->data;
        head<T> = head<T>->next;
    }
}

```

Main Code:

```cpp
#include<iostream>
#include"listStack.h"

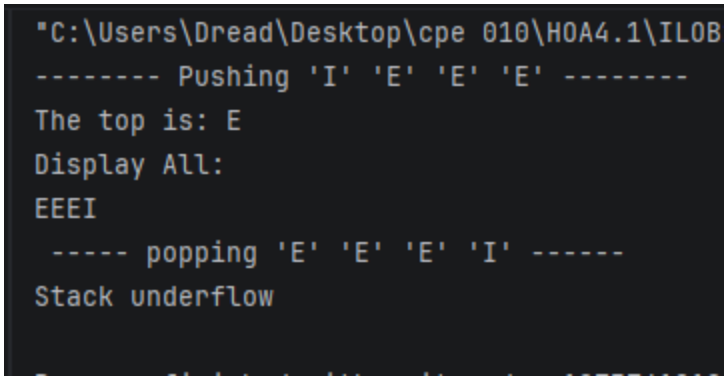
int main(){
    std::cout<<"-------- Pushing 'I' 'E' 'E' 'E' --------\n";

    //pushing
    push( newData: 'I');
    push( newData: 'E');
    push( newData: 'E');
    push( newData: 'E');

    //display top
    Top<char>();
    //display all
    std::cout << "Display All: "<<std::endl;
    displayAll<char>();

    //pop and print each element
    std::cout<<"\n ----- popping 'E' 'E' 'E' 'I' ------\n";
    while(!isEmpty<char>()){
        std::cout<<"popping: "<<pop<char>()<<std::endl;
    }
    // test stack underflow
    pop<char>();
    return 0;
}
```

OUTPUT:



CODE ANALYSIS ILOB2

On the header file, I created a linked list type of stack. I did this by using the template<typename T> which means that T acts as a placeholder for a data type. After that I created the Node class, and then a pointer tip that always points to the current top node, and a global pointer declaration. I then created the pushing operation, and isEmpty() wherein it checks if the stack is empty. Next is a pop() method wherein it removes the Top element of the stack. And then Top() wherein it displays what the top element is, and lastly is the displayAll() wherein it displays all the elements in the stack. Now in the main file, I include iostream and also the header file that I created. Next in the main function I push 4 new character data, and then I proceed to display the top by calling the Top function, and then display each element. After that I pop each element, and lastly I tested the stack underflow.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C

Code in stack using Array:

```cpp
#include <iostream>
#include <string>

const int maxCap = 100;
char stack[maxCap];
int top = -1;

//prototyping
void push(char c);
char pop();
char Top();
bool isFull();
bool isEmpty();
bool isMatching(char a, char b);
bool balanceCheck(const std::string& expression);


int main(){
    std::string inpExpression;
    std::cout << "Balance Checker (Input halt to exit)" << std::endl;

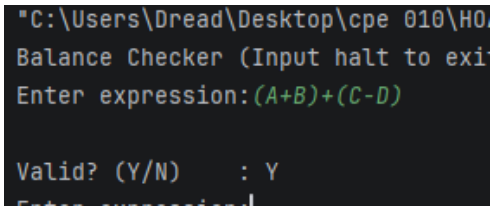
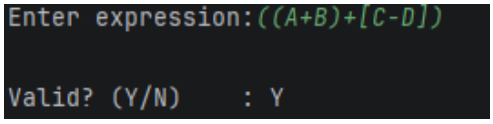
    while (true) {
        std::cout<<"Enter expression: ";
        std::getline( [&]std::cin, [&]inpExpression);
        if (inpExpression == "" || inpExpression == "halt") break;

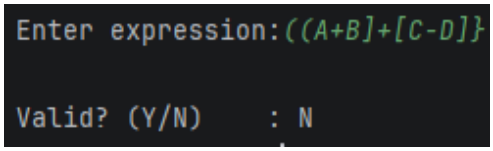
        top = -1;
        bool isValidExpression = balanceCheck(inpExpression);
        std::cout << "Valid? (Y/N)    : " << (isValidExpression ? "Y" : "N") << "\n";
    }
    return 0;
}
```

```
33
34  bool isMatching(char a, char b) {
35      if (a == '(' && b == ')') return true;
36      if (a == '{' && b == '}') return true;
37      if (a == '[' && b == ']') return true;
38      return false;
39  }
40
41  bool balanceCheck(const std::string& expression) {
42      for (char ch : expression) {
43          if (ch == '(' || ch == '{' || ch == '[') {
44              push(ch);
45          }
46          else if (ch == ')' || ch == '}' || ch == ']') {
47              if (isEmpty()) return false;
48              char topChar = pop();
49              if (!isMatching( a: topChar,  b: ch)) {
50                  return false;
51              }
52          }
53      }
54      return isEmpty();
55  }
56
57  bool isEmpty(){
58      return top == -1;
59  }
60
61  bool isFull(){
62      return top >= maxCap - 1;
63  }
64
65  void push(char c){
66      if (isFull()) {
67          std::cout << "Stack Overflow!\n";
68          return;
69      }
70      stack[++top] = c;
71  }
72
```

```
73  char pop(){
74      //check if there are no more elements inside the stack
75      if(isEmpty()) return '\0';
76      //decrement top
77      return stack[top--];
78  }
79
80  char Top(){
81      //error checking
82      if(isEmpty()) return '\0';
83      //display the top element
84      return stack[top];
85  }
86
```

A. Stack using Array

Expression	Valid? (Y/N)	Output (Console Screenshot)	Analysis
(A+B)+(C-D)	Y		The expression is balanced since the program pushes (into the stack for (A+B), and then pops it when it detects). Then the same goes with (C-D).
((A+B)+(C-D)	N		The expression is not valid since it has an unclosed parenthesis in the outer part. Hence the program pushing two (characters, and then pop one after (A+B). And then pushes and pop another one when it comes to (C-D) but after that there is one mismatched because isEmpty() returns false.
((A+B)+[C-D])	Y		This is valid because the opening symbols which are (and [are pushed and their matching symbol is popped thus each character matches its respective

			closing pair.
((A+B]+[C-D]}	N		The expression is not valid because there are a lot of mismatches which cause isMatching() to return false.

Code in stack using Linked Lists:

Header File:

```cpp
#ifndef LIST_STACK_H
#define LIST_STACK_H


template <typename T>
class Node{
    public:
    T data;
    Node<T>* next = nullptr;
    Node(T val) : data(val), next(nullptr) {}
};

template <typename T>
class ListStack {
private:
    Node<T>* head;

public:
    ListStack() : head(nullptr) {}

    ~ListStack() {
        clear();
        }

    bool isEmpty() const {
        return head == nullptr;
    }

    void push(T newData) {
        Node<T>* newNode = new Node<T>(newData);
        newNode->next = head;
        head = newNode;
    }

```

```
35      T pop() {
36          if (isEmpty()) {
37              std::cerr << "Stack Underflow\n";
38              return T();
39          }
40          Node<T>* temp = head;
41          T tempVal = temp->data;
42          head = head->next;
43          delete temp;
44          return tempVal;
45      }
46
47      T peek() const {
48          if (!isEmpty()) return head->data;
49          return T();
50      }
51
52      void clear() {
53          while (!isEmpty()) {
54              pop();
55          }
56      }
57  };
58
59
60  #endif  #ifndef LIST_STACK_H
```

Main File:

```cpp
#include <iostream>
#include <string>
#include"Supplementary_listStack.h"

bool isMatching(char a, char b);
bool balanceCheck(const std::string& expression);

int main() {
    std::string inpExpression;
    std::cout << "Balance Checker (Input halt to exit)" << std::endl;

    while (true) {
        std::cout<<"Enter expression: ";
        std::getline( [&] std::cin, [&] inpExpression);
        if (inpExpression == "" || inpExpression == "halt") break;

        bool isValidExpression = balanceCheck(inpExpression);
        std::cout << "Valid? (Y/N)    : " << (isValidExpression ? "Y" : "N") << "\n";
    }

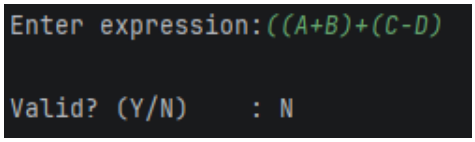
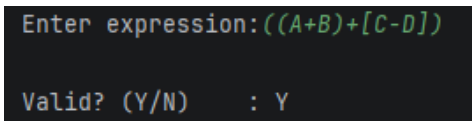
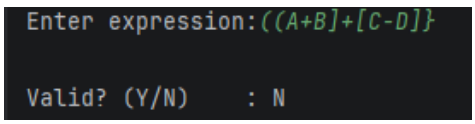
    return 0;
}

bool isMatching(char a, char b) {
    if (a == '(' && b == ')') return true;
    if (a == '{' && b == '}') return true;
    if (a == '[' && b == ']') return true;
    return false;
}
```

```
30
31  bool balanceCheck(const std::string& expression) {
32      ListStack<char> stack;
33      for (char ch : expression) {
34          if (ch == '(' || ch == '{' || ch == '[') {
35              stack.push(ch);
36          }
37          else if (ch == ')' || ch == '}' || ch == ']') {
38              if (stack.isEmpty()) return false;
39              char topChar = stack.pop();
40              if (!isMatching( a: topChar,  b: ch)) {
41                  return false;
42              }
43          }
44      }
45      return stack.isEmpty();
46  }
```

B. Stack using Linked Lists

Expression	Valid? (Y/N)	Output (Console Screenshot)	Analysis
(A+B)+(C-D)	Y	Enter expression: *(A+B)+(C-D)* Valid? (Y/N) : Y	The stack dynamically allocates a node for (when it has been pushed and also deallocates it during pop when it matches its closing part. This is being repeated for the other sub expression which is (C-D). Now when reaching the end of the input the head is pointed to nullptr since isEmpty() is true thus it being valid.

((A+B)+(C-D)	N		This expression is not valid because two opening parenthesis are pushed into the linked list stack, since the extra opening parenthesis have no partner closing parentheses in the popping procedure hence isEmpty is returning false.
((A+B)+[C-D])	Y		The expression is valid because all opening symbols that have been pushed have a corresponding closing symbol that have been popped, and also both have been compared by isMatching() method. Hence the head being pointed to nullptr.
((A+B]+[C-D]}	N		This is not valid because the isMatching condition fails immediately and returns false. Hence, the destructor is called and deallocates any unpopped nodes.

Tools Analysis:

- How do the different internal representations affect the implementation and usage of the stack?
    - In a fixed size array representation the size is fixed thus contiguous memory is allocated upfront making memory allocation predictable and operations fast. While in linked list representation every push dynamically allocates a new node on the heap thus resulting in heap allocated nodes being scattered across the memory, which means that cache performance is poor.

Conclusion:

In this activity I learned about stacks, and how to create a stack using C++ STL. Moreover, I developed C++ code that uses both arrays and linked lists to create a stack, and solve problems using an implementation of stack. Furthermore, I also did an analysis for every procedure and supplementary activity. Lastly, this activity was very challenging especially the supplementary part, but alas I have completed it. With that, I think that I still have a lot to improve and I will keep learning to enhance my programming skills.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Concept explanation, and code assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used AI as a tool for understanding and guide.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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